Jack **Lafferty**

MECHATRONICS ENGINEER • ROBOTICS & AUTOMATION

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PROFILE

Mechatronics engineer who spent 14 months as the sole hardware engineer at a heat pump controls startup, owning its core product from prototype to production: a redesign for volume manufacture (-36% unit cost) and an automated QC system used on 4,000+ devices. Final year MEng student at Imperial College London, building ROS 2 and computer vision systems, seeking a graduate robotics role.

EDUCATION

Imperial College London *2022 – 2027*

MEng Mechanical Engineering with a Year in Industry

- Predicted 2:1, with >70% in all project work; graduating summer 2027
- Final year project: developing a camera module with ML defect detection for a robotic battery cell line
- Robotic manipulators project: developing control software for a robotic arm and gripper
- Relevant modules: Robotics & Automation, Mechatronics, Machine Dynamics

EXPERIENCE

Adia Thermal, London *Aug 2025 – Present*

Mechatronics Engineering Intern (full-time), then Junior Mechatronics Engineer (part-time from Oct 2026)

Sole hardware engineer at a 10-person heat pump startup making the Adia Hub (pump, sensors, Pi/PCBA controller).

- Led a total redesign of the core product for volume manufacture, cutting unit cost by 36% (from loss-making to margin-positive) and halving assembly time, with tooling complete. Developed a further concept at 73% lower cost.
- Ran component testing from specification to selection: defined requirements, sourced suppliers and samples, set acceptance criteria, built test rigs, wrote control code, analysed data and presented recommendations to the founders.
- Wrote and optimised the QC software and pass criteria with the software team: a Raspberry Pi guides inspection, calibrates each device via its PIC18 and logs results to a dashboard. Used on 4,000+ devices, cutting the failure rate from 11% to <1% by root-causing a firmware fault and an uncaught motor defect.
- Designed custom PCBs in KiCad (made by PCBWay) to eliminate tedious manual production tasks.
- Established a supply chain across China, India, Poland and the UK, from RFQs through to production deadlines.

Earlier work: construction labourer (self-employed, summers 2019–2024) and mechanics apprentice at Southbourne Park Garage (summer 2021), Bournemouth.

PROJECTS

Automatic Wire Cutter and Stripper *2026*

- Built a wire cutting and stripping machine for low-volume production (Raspberry Pi Pico, two steppers, 3D printed feed and cut mechanisms), cutting a recurring 3-hour task to 30 minutes with more accurate cuts.

Vision-Guided Pick-and-Place Arm *2026 – present*

- ROS 2 pick-and-place system: an OpenCV node locates blocks from an RGB-D camera and MoveIt plans inverse-kinematics grasps. Currently at the simulation stage (Gazebo).

DC Motor Position Control (Simulink + PID) *2026 – present*

- Simulink model of a DC motor and a PID position controller written in C on a Raspberry Pi Pico with encoder feedback; currently validating the model against the hardware.

Vision-Guided Maze Robot *2026 – present*

- Autonomous vehicle programmed in embedded C that navigates a maze using visual cues.

Formula Student, Imperial College London *2024 – 2025*

- Suspension designer and project manager. Designed, analysed (CAD, FEA) and prototyped the car's novel suspension system, part of a 10.1% vehicle mass reduction, while running the team's schedule and deliverables.

Independent projects: built a DC motor from scratch, restored an outboard boat engine and repaired a motorcycle.

SKILLS

CAD and analysis: Fusion 360, SolidWorks, ANSYS (FEA)

Electronics: Raspberry Pi, PIC microcontrollers, PCB design (KiCad), sensors and actuators, test rigs

Software: Python, C, MATLAB, Git (branching, pull requests), Docker, command line and SSH, custom Raspberry Pi images

Robotics and controls: ROS 2, MoveIt and Gazebo (simulation), OpenCV, Simulink, PID control

Manufacturing: design for manufacture and assembly, supplier management, quality control

Fabrication: lathe, mill, waterjet, 3D printing; designed for laser cutting, sheet metal, vacuum forming, pipework

Interests: Films (650+ reviewed), hiking (recently wild camping on Dartmoor), chess